# 36 · Oestrogen, progesterone and the pill

The words of this track, as it says them. Bold brown is the examiner.

**Oestrogen, progesterone, and the pill. Five marks.**

**Describe the physiological effects of oestrogen and progesterone, including the female secondary sex characteristics, and explain how the contraceptive pill prevents pregnancy.**

The hook: oestrogen builds, and progesterone prepares. The pill works three ways: no egg, a mucus plug, and a thin lining.

Oestrogen, from the granulosa cells of the follicle, rises at puberty. It promotes oogenesis and follicle growth, and grows the uterine tubes, the uterus and the vagina to adult size. Its secondary sex characteristics: breast growth, fat laid down on the hips and breasts, a wider, lighter pelvis, and axillary and pubic hair. And in each cycle, oestrogen rebuilds the endometrium, in the proliferative phase.

Progesterone, from the corpus luteum, prepares the endometrium for implantation, in the secretory phase. And in pregnancy, it develops the breast lobules and alveoli.

The pill contains synthetic oestrogen and progestin, which is a synthetic progesterone, either combined, or progestin alone. It prevents pregnancy three ways. It stops ovulation. It thickens the cervical mucus, so sperm can't get into the uterus. And it thins the endometrium, so a fertilised egg can't implant.

The traps. Oestrogen: the ducts, the fat, the pelvis, the proliferative phase. Progesterone: the alveoli, the secretory phase, pregnancy. Second: the pill works three ways, so name all three for full marks. Third: progestin is the synthetic form of progesterone.

**Once more. Oestrogen, progesterone, and the pill.**

Oestrogen rises at puberty and drives follicle growth. It grows the uterine tubes, uterus and vagina; it brings breast growth, fat on the hips and breasts, a wider, lighter pelvis, and axillary and pubic hair; and it rebuilds the endometrium in the proliferative phase. Progesterone, from the corpus luteum, readies the lining for implantation, and in pregnancy develops the lobules and alveoli. The pill, synthetic oestrogen and progestin, stops ovulation, thickens the cervical mucus, and thins the endometrium.